

Q.No:4

Question description

Byteland has n cities and m roads between them. The goal is to construct new roads so that there is a route between any two cities. Your task is to find out the minimum number of roads required, and also determine which roads should be built.

Constraints

$1 \leq n \leq 10^5$
 $1 \leq m \leq 2 \cdot 10^5$
 $1 \leq a, b \leq n$

Input

The first input line has two integers n and m : the number of cities and roads. The cities are numbered $1, 2, \dots, n$.

After that, there are m lines describing the roads. Each line has two integers a and b : there is a road between those cities.

A road always connects two different cities, and there is at most one road between any two cities.

Output

First print an integer k : the number of required roads.

Then, print k lines that describe the new roads. You can print any valid solution.

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN) 4 2 1 2 3 4	INPUT (STDIN) 4 2 2 3 2 4
EXPECTED OUTPUT 1 2 4	EXPECTED OUTPUT 1 1 4

Mandatory Test Cases

Test Case 1
KEYWORD while(m--)

Code:

```
#include <stdio.h>

#define MAX 100005

int parent[MAX];
int size[MAX];

int find(int x)
```

```
{
while (parent[x] != x)
{
parent[x] = parent[parent[x]];
x = parent[x];
}
return x;
}

void unite(int a, int b)
{
int pa = find(a);
int pb = find(b);

if (pa == pb)
return;

if (size[pa] < size[pb])
{
int temp = pa;
pa = pb;
pb = temp;
}

parent[pb] = pa;
size[pa] += size[pb];
}

int main()
{
int n, m;
scanf("%d %d", &n, &m);

for (int i = 1; i <= n; i++)
{
parent[i] = i;
size[i] = 1;
}

while (m--)
{
int a, b;
scanf("%d %d", &a, &b);

unite(a, b);
}

int representative[MAX];
int count = 0;

printf("Output:\n");

for (int i = 1; i <= n; i++)
{
```

```
if (find(i) == i)
{
    representative[count] = i;
    count++;
}
}

printf("%d\n", count - 1);

for (int i = 1; i < count; i++)
{
    printf("%d %d\n", representative[0], representative[i]);
}

return 0;
}
```

Output:

Test 1:

```
4 2
1 2
3 4
Output:
1
1 3
aadityajaiswal@Aadity
```

Test 2:

```
4 2
2 3
2 4
Output:
1
1 2
aadityajaiswal@Aadity
```